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Note on the inversion theorem

By J. GIL-PELAEZ

If $F(x)$ is a one-dimensional distribution function, its corresponding characteristic function is the complex function of the real variable t:

$$\phi(t) = \int_{-\infty}^{\infty} e^{itx}\, dF(x).$$

The theorem which proves that the distribution is uniquely determined by the characteristic function is called the inversion theorem. This was first demonstrated by Lévy, who found the expression

$$F(x) - F(0) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \frac{1-e^{-itx}}{it}\phi(t)\, dt.$$

The constant $F(0)$ is no obstacle to an ascertainment of the distribution, because by subtraction we have for every interval (a, b)

$$\int_a^b dF(x) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \frac{e^{-ita}-e^{-itb}}{it}\phi(t)\, dt.$$

Furthermore, when the distribution belongs to the continuous type the constant is also eliminated by differentiation,

$$f(x) = F'(x) = \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-itx}\phi(t)\, dt.$$

This is sufficient in the majority of cases, but when we are interested in the explicit expression of the distribution function, the constant $F(0)$ is a substantial inconvenience. I therefore present a derivation of the inversion theorem by a new method which has the advantage of evaluating the constant term. I shall, in fact, show that

$$F(x) = \frac{1}{2} + \frac{1}{2\pi}\int_0^{\infty} \frac{e^{itx}\phi(-t) - e^{-itx}\phi(t)}{it}\, dt.$$

In order to prove this, let us consider the step-function

$$\operatorname{sign}(y-x) = \frac{2}{\pi}\int_0^{\infty} \frac{\sin t(y-x)}{t}\, dt = \begin{cases} -1 & \text{if} \quad y < x, \\ 0 & \text{if} \quad y = x, \\ 1 & \text{if} \quad y > x, \end{cases}$$

and observe that

$$\int_{-\infty}^{\infty} \operatorname{sign}(y-x)\, dF(y) = 1 - 2F(x).$$

Now, for any positive numbers ϵ, λ, we have

$$\frac{1}{\pi}\int_{\epsilon}^{\lambda} \frac{e^{itx}\phi(-t) - e^{-itx}\phi(t)}{it}\, dt = \frac{1}{\pi}\int_{\epsilon}^{\lambda}\int_{-\infty}^{\infty} \frac{e^{-it(y-x)} - e^{it(y-x)}}{it}\, dF(y)\, dt$$

$$= -\frac{2}{\pi}\int_{\epsilon}^{\lambda}\int_{-\infty}^{\infty} \frac{\sin t(y-x)}{t}\, dF(y)\, dt = -\int_{-\infty}^{\infty} dF(y)\frac{2}{\pi}\int_{\epsilon}^{\lambda} \frac{\sin t(y-x)}{t}\, dt.$$

The inversion of the order of integration has been possible because for $\epsilon < t < \lambda$ and $-\infty < y < \infty$

$$\left| \frac{\sin t(y-x)}{t} \right| < \frac{1}{\epsilon}.$$

When ϵ tends to zero and λ tends to infinity, we have

$$\frac{1}{\pi}\int_0^{\infty} \frac{e^{itx}\phi(-t) - e^{-itx}\phi(t)}{it}\, dt = -\int_{-\infty}^{\infty} dF(y) \lim \frac{2}{\pi}\int_{\epsilon}^{\lambda} \frac{\sin t(y-x)}{t}\, dt$$

$$= -\int_{-\infty}^{\infty} \operatorname{sign}(y-x)\, dF(y) = 2F(x) - 1,$$

which establishes the theorem. We observe that it is possible to take the limit inside the integral sign because

$$\int_{\epsilon}^{\lambda} \frac{\sin t(y-x)}{t}\, dt$$

is a continuous function of ϵ, λ with a bounded modulus.

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When the characteristic function is of the following kind,

$$\phi(t) = \sum_r p_r e^{itx_r},$$

the formula of the inversion theorem gives us the following function:

$$F(x) = \frac{1}{2} + \frac{1}{2\pi}\int_0^\infty \frac{\sum_r p_r e^{it(x-x_r)} - \sum_r p_r e^{-it(x-x_r)}}{it}\,dt$$

$$= \frac{1}{2} + \sum_r p_r \frac{1}{\pi}\int_0^\infty \frac{\sin t(x-x_r)}{t}\,dt = \frac{1}{2} + \frac{1}{2}\sum_r p_r \operatorname{sign}(x-x_r).$$

This is a step-function with a saltus p_r at x_r. We must observe that at the points of discontinuity the value of the function is the half-sum of the limits on the right and on the left. This is not the value if the distribution function is taken to be continuous on the left, but depends on the Dirichlet integral which has been used in the proof.

I have to thank Mr D. G. Kendall for his helpful suggestions.

A note on the statistical independence of quadratic forms in the analysis of variance

By J. A. NELDER, *National Vegetable Research Station*

A widely used notation in writing out the algebraic identities in the analysis of variance is that employing multiple suffixes and using dots to denote averaging over the suffices concerned. Thus with $p \times q$ two-way classification (to take a simple example) we have

$$\sum_{i,j} x_{ij}^2 \equiv \sum_{i,j} x^2_{..} + \sum_{i,j}(x_{i.}-x_{..})^2 + \sum_{i,j}(x_{.j}-x_{..})^2 + \sum_{i,j}(x_{ij}-x_{i.}-x_{.j}+x_{..})^2. \tag{1}$$

Let us denote the four linear forms in the x_{ij}, namely,

$$x_{..},\quad x_{i.}-x_{..},\quad x_{.j}-x_{..}\quad \text{and}\quad x_{ij}-x_{i.}-x_{.j}+x_{..}$$

by

$$y_{1ij},\quad y_{2ij},\quad y_{3ij}\quad \text{and}\quad y_{4ij}$$

respectively. Then writing x, y_1, y_2, y_3, y_4 for the five $pq \times 1$ vectors

$$\{x_{ij}\},\ \{y_{1ij}\},\ \{y_{2ij}\},\ \{y_{3ij}\},\ \{y_{4ij}\}$$

and putting $y_\alpha = B_\alpha x$ we note that

(i) B_α is symmetrical for all α,

(ii) $B_1+B_2+B_3+B_4 = I$, since

$$x_{ij} \equiv x_{..} + (x_{i.}-x_{..}) + (x_{.j}-x_{..}) + (x_{ij}-x_{i.}-x_{.j}+x_{..}),$$

(iii) $B_\alpha B_\beta$ is skew symmetric, $\alpha \neq \beta$, since each cross-product such as

$$\sum_{i,j}(x_{.j}-x_{..})(x_{ij}-x_{i.}-x_{.j}+x_{..}) = x'B_3B_4x$$

vanishes identically in the x_{ij}.

Now, although accounts of the analysis of variance note the separate vanishing of each of the cross-products in (1) and similar identities, it does not seem to have been pointed out that when the x_{ij} are independent normal variates with zero mean and unit variance, the conditions (i), (ii) and (iii) above on the B matrices are sufficient for the quadratic forms on the right-hand side of (i), i.e.

$$(B_\alpha x)' B_\alpha x = x'B_\alpha' B_\alpha x = x'B_\alpha^2 x \quad (\alpha = 1, 2, 3, 4),$$

to be independently distributed as χ^2 and that the number of degrees of freedom associated with the form $x'B_\alpha^2 x$ is trace B_α. Thus the number of degrees of freedom of $\sum_{i,j}(x_{ij}-x_{i.}-x_{.j}+x_{..})^2$ is given by

$$\text{trace } B_4 = pq\left(1-\frac{1}{q}-\frac{1}{p}+\frac{1}{pq}\right) = pq-p-q+1,$$

a familiar result.

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